Bhavinkumar Umatiya

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Summary

Final-year ECE student in FPGA/hardware acceleration and RTL design: fully pipelined Vitis HLS accelerators on Xilinx ZCU104 (300 MHz closure, 50.7x over ARM) and CDC-safe RTL verified with a complete UVM 1.2 testbench (constrained-random, SVA, functional coverage). Merged upstream contributions to Zephyr RTOS, ROS 2 and BlackParrot RISC-V. **Available for a 6-month internship (final-year, Nirma University).**

Technical Skills

Languages: C, C++, Python, Verilog, SystemVerilog, Tcl scripting, Bash, ARM/RISC-V Assembly
Verification: UVM 1.2, OO testbench design (SystemVerilog/C++), SystemVerilog Assertions (SVA), Functional Coverage, Constrained-Random, Scoreboards
FPGA / EDA: Vivado, Vitis HLS, Vitis AI (DPU), XSim, ModelSim/Questa, Quartus, Verilator, MATLAB, Git
Platforms: Xilinx Zynq UltraScale+ (ZCU104), Altera DE2, STM32, ESP32, Teensy, Nordic nRF (BLE)
Protocols / Concepts: AXI4/AXI-Lite/AXI-Stream, CDC & Gray-code sync, STA/timing closure, SPI/I2C/UART, BLE, IEEE 802.15.4

Projects

Heterogeneous ViT Inference Accelerator | GitHub — May 2026
Xilinx ZCU104 · Vitis HLS · Vitis AI DPU · AXI
- Architected a PS/PL partitioned Vision-Transformer accelerator: Softmax, GELU and LayerNorm offloaded from the Cortex-A53 to custom fully pipelined HLS IP over AXI, with the matmul path on the DPU.
- Closed timing at 300 MHz using PIPELINE/DATAFLOW pragmas and interface optimisation; measured 72.5 ms end-to-end inference vs 3671 ms on the ARM baseline (50.7x speedup, 15.4 W board power).

Adaptive ROI-Based Edge-AI Video Pipeline | GitHub — May 2026
Xilinx ZCU104 · Vitis AI DPU (YOLOv4) · VCU H.264
- Implemented an on-device surveillance pipeline: YOLOv4 on the DPU drives 3-zone ROI priority masking feeding hardware H.264 encoding on the Video Codec Unit – no host-side processing.

Asynchronous FIFO with UVM Verification | GitHub — Apr 2026
Verilog RTL · SystemVerilog · UVM 1.2 · SVA
- Designed a parameterised CDC-safe asynchronous FIFO with Gray-coded pointers and 2-stage synchronisers; verified full/empty, overflow/underflow and simultaneous read/write corner cases.
- Built a complete UVM testbench – sequencer/driver/monitor, constrained-random stimulus, SVA assertions, cross functional coverage and a reference-model scoreboard; 500-transaction constrained-random regression plus directed overflow/underflow tests, all passing.

Open Source Contributions — merged upstream

BlackParrot RISC-V Core – Bespoke Silicon Group / FOSSi Foundation — 2026
Merged documentation fix; full simulation bring-up
- Built and validated the BlackParrot RISC-V simulation environment end-to-end with Verilator to exercise the instruction pipeline; corrected architectural documentation upstream.

Zephyr RTOS – The Linux Foundation — Feb 2026 – Present
Upstream contributor – 3 PRs merged to mainline (#106403, #106404, #106472)
- Fixed the Nordic Bluetooth Link Layer passing raw microsecond values to the hardware timer (nrf_timer_cc_set) by introducing a tick-conversion define, eliminating BLE scheduling assertion failures (PR #106472).

ROS 2 Controllers – Open Robotics — Feb 2026 – Present
3 merged PRs (ros2_ controllers)
- Refactored the diff-drive controller: removed a deprecated parameter, migrated exception-based initialisation to the ros2_control lifecycle API, and resolved a runtime anti-pattern.

Experience

Oizom Instruments Pvt. Ltd. — May 2026 – Jul 2026
Hardware Engineer Intern · Ahmedabad
- Built a Python OCR/computer-vision test-automation system driving Room EQ Wizard acoustic calibration with zero manual GUI interaction, cutting each noise-sensor test from 60 to 30 minutes (3 tests/day, ~1.5 engineer-hours saved daily); interfaced a Teensy MCU over serial for real-time acquisition from Class 1 acoustic sensors.

Education

Institute of Technology, Nirma University – B.Tech ECE (Minor: Data Science) — 2024 – 2027
CGPA 6.53/10 · Ahmedabad

Dalia Institute of Diploma Studies (GTU) – Diploma in IT — 2021 – 2024
CGPA 7.11/10 · Kheda

Achievements & Certifications

- **BharatPart – Founder, side project** (bharatpart.com) – parametric electronic-component search and BOM checker aggregating Mouser, DigiKey and Indian distributors with live INR pricing
- AMD Slingshot – Top 4, AMD India Campus Days 2026
- Smart India Hackathon 2025 – Campus finalist (200+ teams); 1st place, internal hardware hackathon
- BAREMETALSS – YouTube channel on FPGA design, Vivado workflows and hardware acceleration
- Certifications: Semiconductor Packaging & Test (Micron) · Generative AI (Google Cloud) · Azure AI Fundamentals